



Deep Learning IndabaX

South Africa 2026

From Foundations to Frontiers: Africa's AI Journey

6–9 July 2026

● University of KwaZulu-Natal, Westville Campus · Durban

Lecture theatres 2, 4 & 5 are located at the T-block.

SESSION TYPE

● Keynote ● Fundamentals ● Contributed talk ● Workshop ● Hackathon
● Debate / Panel ● Meetup / Event

The annual meeting of the South African machine learning & AI community.



08:00 09:00	Coffee & Registration Registration continues throughout the day, and during each coffee break		
09:00 11:00	SOFT SKILLS WORKSHOP · THEATRE 2 Math for ML	HACKATHON · BEGINNER TRACK · THEATRE 4 Zindi Hackathon Intro Session: Financial Inclusion in Africa	SOFT SKILLS WORKSHOP · THEATRE 5 Science communication
11:00 11:30	Coffee & Snacks		
11:30 12:00	Opening Address <i>Theatre 4</i>		
12:00 13:00	KEYNOTE · THEATRE 4 Dr Ndivhuwo Makondo: <i>Agentic Data Analysis: can LLM Agents Automate Data Analysis?</i>		
13:00 14:00	Lunch		
14:00 15:00	DEBATE · THEATRE 4 Should autonomous AI agents operate independently in real-world systems?		
15:00 15:30	Group Photo		
15:30 19:30	MEETUP / EVENT · MAIN HALL (ACROSS THE POND) AI Fest continues to 19:30		

08:00 09:00	Morning coffee <i>Late registration 08:00 – 08:30</i>			
09:00 10:00	KEYNOTE · THEATRE 4 Prof Okuthe P. Kogeda: <i>AI and Education: Current status, challenges and opportunities</i>			
10:00 10:30	Coffee & Snacks <i>Speaker intros start 10:25</i>			
10:30 11:45	FUNDAMENTALS · THEATRE 4 Machine Learning <i>Randhir Rawatlal</i>	CONTRIBUTED TALK · THEATRE 2 10:30 Building Sustainable Food Systems with AI for Food Waste Reduction (Andronicus A. Akinyelu) 11:00 Integrating Health Behaviour and AI/ML Theories: A Case for Pre-Screening Prediction in Industry-Sponsored Clinical Trials (Janine Zitianiellis) 11:30 When South Africans Ask AI About HIV: Evaluating Safety, Accuracy, and Trust in Large Language Models (Mariam Gustavo)	CONTRIBUTED TALK · THEATRE 5 10:30 The playbook to technological sovereignty (David Phume) 11:00 The Landscape of the Semiconductor Industry in Africa: From Minerals to Microchips (Laing Lourens) 11:30 AI risk, African agency, and the institutions we need (Charl Botha)	
11:45 12:30	DEBATE · THEATRE 5 Is AI truly solving Africa's challenges, or are we overestimating its impact?			
12:30 13:30	Lunch			
13:30 14:30	KEYNOTE PANEL · THEATRE 4 AI Policy			
14:30 15:00	Break <i>Speaker intros start 14:25</i>			
15:00 16:30	FUNDAMENTALS · THEATRE 4 The physics of machine learning <i>Maria Schuld</i>	CONTRIBUTED TALK · THEATRE 2 15:00 Developing AI Guidelines for Infrastructure-as-a-Service Platforms (Binjamin Barsch) 15:30 AI Integration in South African Strategic Communication: Disparities in Graduate Preparedness, Industry Expectations, and Academic Responses (Ibrahim Lukombe) 16:00 AI for Community Intelligence: A Graduate-Led Municipal Data Centre Model for Evidence-Based Local Development in KwaZulu-Natal (Innocent N Mthembu)	MEETUP · THEATRE 1 15:00 RL 15:30 CV 16:00 NLP	HACKATHON · ADVANCED TRACK · THEATRE 5 Hackathon Intro for Advanced Hackathon
16:30 17:00	Coffee & Snacks			
17:00 18:00	CONTRIBUTED TALK · THEATRE 2 17:00 Predicting the Unseen: AI-Driven Kinetic Modeling of Under-Film Corrosion in Nano-Silica/Epoxy Coatings for Acidic Oilfield Environments (Beulah Ude) 17:15 Explainable Malware Detection Using Machine Learning (Samkelo Msibi)			
17:30 midnight	HACKATHON · H1 BUILDING Beginner & Advanced Hackathons Begin 17:30 – midnight			

08:00
09:00

Morning coffee

09:00
10:00

KEYNOTE · THEATRE 4

Dr Herkulaas Combrink: *When Information Spreads Faster Than Truth: Why Computational Infodemiology Matters*10:00
10:30Coffee & Snacks *Speaker intros start 10:25*10:30
11:45

FUNDAMENTALS · THEATRE 4

LLM Agents

Naweed Khan

CONTRIBUTED TALK · THEATRE 2

10:30 Balancing Trade-offs: Why Multi-Objective Reinforcement Learning Matters (Asad Jeewa)

11:00 Techno-Economic, Environmental and Social Multi-Objective Optimisation of Hybrid Renewable Energy Systems using Metaheuristic Algorithms (Dhruti Dheda)

11:30 Algorithm Design and Mathematical Modeling for Efficient Automatic Speech Recognition in Low-Resource African Languages (Simanga Comfort Mchunu)

CONTRIBUTED TALK · THEATRE 5

10:30 Applied AI for Urban Governance: Practical Use Cases from Open Cities Lab (Wasim Moosa)

11:00 Building AI Ecosystems for African Agriculture: Lessons from Scaling OptiGrow to 6,000+ Users (Baraka Kileo)

11:30 Text-to-Speech for African languages: Commercialisation and opportunities (Aby Louw)

11:45
12:30

DEBATE · THEATRE 2

Can Africa achieve AI sovereignty, or will it remain dependent on global AI powers?

12:30
13:30

Lunch

13:30
14:30

WORKSHOP · THEATRE 2

Computational Infodemiology and Retrieval-Augmented Generation (RAG) with Knowledge Graphs

WORKSHOP · THEATRE 5

Responsible AI and Computing: What Is Your Take?

14:30
15:00Break *Speaker intros start 14:25*15:00
16:30

WORKSHOP · THEATRE 2

Computational Infodemiology and RAG with Knowledge Graphs

WORKSHOP · THEATRE 5

Responsible AI and Computing: What Is Your Take?

16:30
17:00

Coffee & Snacks

17:00
18:00

FLASH TALKS · THEATRE 4

Flash Talks (12) ≈ 1 hour

17:30
midnight

HACKATHON · H1 BUILDING

Beginner & Advanced Hackathon Continue: Submit by Midnight! 17:30 – midnight

08:00
09:00

Morning coffee

09:00
10:00

KEYNOTE · THEATRE 4

Prof Chijioke Okorie: *All-hands-on-deck AI governance*

10:00
10:30

Coffee & Snacks *Speaker intros start 10:25*

10:30
12:00

FUNDAMENTALS · THEATRE 4

Computer Vision

Oliver Welsh

WORKSHOP · THEATRE 2

Building Practical AI Agents with
LangGraph and LangChain
Fundamentals

WORKSHOP · THEATRE 5

AI Safety and Security in African
Contexts: From Governance to
Jailbreak

12:00
13:00

Lunch

13:00
14:00

CONTRIBUTED TALK · THEATRE 4

13:00 Examining the applicability of the
Protection of Personal Information Act in AI-
driven environments (Vicent Mbonye)

13:30 Self-Supervised Vision Transformers
for Geometric 3D Reconstruction and
Biomass Estimation (Aaron Masuba)

WORKSHOP · THEATRE 2

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14:00
14:30

Break *Speaker intros start 14:25*

14:30
15:30

KEYNOTE · THEATRE 4

Dr Herkulaas Combrink: *When
Information Spreads Faster Than Truth:
Why Computational Infodemiology
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15:30
16:00

Coffee & Snacks

16:00
16:30

Prize Giving and Closing Ceremony *Theatre 4*